Int. J. Production Economics 135 (2012) 37–49

Contents lists available at ScienceDirect

Int. J. Production Economics

journal homepage: www.elsevier.com/locate/ijpe

Design of sustainable supply chains under the emission trading scheme

A. Chaabane*, A. Ramudhin, M. Paquet

École de technologie supérieure, Canada

ARTICLE INFO

Article history:
Received 30 November 2009
Accepted 23 October 2010
Available online 28 October 2010

Keywords:
Sustainable supply chain design
Environment
Greenhouse gases emissions
Life cycle assessment
Reverse logistics
Emission trading scheme
Carbon management

ABSTRACT

Increase in environmental concerns together with legislations are forcing industries to take a fresh look at the impact of their supply chain operations on the environment. This paper introduces a mixed-integer linear programming based framework for sustainable supply chain design that considers life cycle assessment (LCA) principles in addition to the traditional material balance constraints at each node in the supply chain. Indeed, the framework distinguishes between solid and liquid wastes, as well as gaseous emissions due to various production processes and transportation systems. The framework is used to evaluate the tradeoffs between economic and environmental objectives under various cost and operating strategies in the aluminum industry. The results suggest that current legislation and Emission Trading Schemes (ETS) must be strengthened and harmonized at the global level in order to drive a meaningful environmental strategy. Moreover, the model demonstrates that efficient carbon management strategies will help decision makers to achieve sustainability objectives in a cost-effective manner.

© 2010 Elsevier B.V. All rights reserved.

1. Introduction

Supply chain network design attempts to define the best supply chain configuration that enables an organization to maximize its long-term economic performance. Typically, the decisions cover two planning levels: (1) strategic decisions on sourcing, production (opening or closing of facilities), distribution and sales; (2) tactical decisions on supply network planning affecting the flow of goods trough the network. Flexibility, robustness, and responsiveness are some of the strategies that have been used to adapt to dynamic changes in the supply chain environment (Sabri and Beamon, 2000). But, unfortunately the pursuit of short term profitability is still recognized as the one of the major drivers for managerial decisions and this, among other things, has contributed to the slowdown in the current global economy.

Nowadays, given the constraints relative to the availability of non-renewable resources (metal, oil, etc.), enterprises are more than ever obliged to rethink their strategies to ensure the sustainability of their operations. Closed-loop supply chains are one of the options that are being considered (Lieckens and Vandaele, 2007; Barker and Zabinsky, 2008; Srivastava, 2008; Pochampally et al., 2009). Other avenues being studied include different actions related to one or more phases of the product life cycle such as product design (Hugo and Pistikopoulos, 2005), production planning and control for remanufacturing (Jayaraman et al., 1999; Luo et al., 2001), inventory management (Ferretti et al., 2007), product recovery (Jayaraman, 2006), reverse logistics (Sheu et al., 2005; Sheu, 2008) and carbon emissions reduction (Ramudhin et al., 2008).

However, these actions may not be enough to guarantee long-term sustainability. Indeed, recovery of used products and re-processing (remanufacturing, recycling, disposal, incineration, etc.) might not only increase operating costs but also contribute to an increase in greenhouse gases (GHG) emissions which defeats long-term stainability. Sustainable development recognizes the interdependence between three dimensions: the economic, the environmental, and the social performances of an organization. An integrated approach that links supply chain decisions to the three pillars of sustainability is advocated.

Sustainable supply chain design (Frota Neto et al., 2008) is a new emerging approach that arose in response to this situation and tries to embed economic, environmental as well as societal decisions in supply chains at design time. The objective of the methodology proposed in this paper is to present a formal decision model that considers the important dimensions of sustainability throughout the supply chain life cycle.

2. Literature review

Traditionally, the main objective of optimization models used in strategic network design focused on the economic aspect of supply chains (Goetschalcks and Fleischmann, 2008). However, more recently there has been a growing awareness about environmental issues. The first proposals tried to integrate such considerations at the plant level. The main drawback of these approaches is that it may result in solutions that reduce the negative environmental

* Correspondence to: 1100, Notre Dame Street Ouest, Montreal, Quebec, Canada H3C 1K3.
E-mail address: amin.chaabane@etsmtl.ca (A. Chaabane).

0925-5273/$ - see front matter © 2010 Elsevier B.V. All rights reserved.
doi:10.1016/j.ijpe.2010.10.025

impact somewhere in the supply chain at the expense of increasing it elsewhere.

Life Cycle Assessment (LCA) methodology has been proposed in response to this situation (De Benedetto and Klemes, 2009). LCA is a process for evaluating the environmental impacts associated with a product, process or activity. It identifies and quantifies the energy and materials used and the waste released to the environment, and evaluates and implements opportunities for environmental improvements. The assessment covers the entire life cycle of the product, process or activity, including extracting and processing raw materials, manufacturing, transportation and distribution, reuse and maintenance, recycling and final disposal.

Hugo and Pistikopoulos (2005) present a mathematical programming-based methodology with explicit inclusion of life cycle assessment (LCA) criteria as part of the strategic investment decisions related to the design and planning of supply chain networks. Nagurney et al. (2006) develop a supply chain model in which the manufacturers can produce homogeneous product in different manufacturing plants with distinct environmental emissions. Frota Neto et al. (2008) develop a framework for the design and evaluation of sustainable logistic networks where activities affecting the environment and cost efficiency in logistic networks are considered. Guillen-Gosalbez and Grossmann (2009) present a supply chain network design model to determine the supply chain configuration along with the planning decisions that maximizes the net present value and minimizes environmental impact. The model includes structural and planning decisions.

Suitable performance measures in evaluating the supply chain are important and directly affect their applicability (Gunasekaran et al., 2004). Various types of performance measures have been used to evaluate sustainable supply chain. Most frequently, they combine the economic performance with the environmental performance in order to find the trade-off between the two performances (Nagurney and Toyasaki, 2003; Pistikopoulos and Hugo, 2005; Sheu et al., 2005; Lu et al., 2007; Frota Neto et al., 2008; Guillen-Gosalbez and Grossmann, 2009). The economic dimension represents the cost or the profit in net present value (Pistikopoulos and Hugo, 2005). Various performance metrics have been developed to evaluate quantitatively the environmental impact of products, processes and activities such as the emissions of GHG (CO_2, CFC, NO*x*, …) (Luo et al., 2001), waste generation (liquid or solid), energy use, and material recovery.

In recent years, different comprehensive environmental performance metrics has been proposed such as the "Eco-indicator 95" (Brentrup et al., 2001) "Eco-indicator 99" (Contreras et al., 2009), "Ecological Footprints", and "EcoPro" (Luo et al., 2001). These metrics are based on different methodological structures and weighting techniques where assumptions are different.

The applicability of different supply chain models have been tested in real industrial cases and in different fields: petrochemical production (Zhou et al., 2000), aluminum industry (Ferretti et al., 2007), personal computer (Min and Melachrinoudis, 1999; Dotoli et al., 2005, 2006), and the pulp and paper industry (Frota Neto et al., 2008). It shows particularly that numerous initiatives have provided incentives for organizations to become more sustainable. Some of these regulations are mandatory, but increasingly others are just voluntary environmental programs and considered as new alternatives for gaining or maintaining a competitive advantage. For instance, many industries are engaged in voluntary RL activities like the automotive industry (Schultmann et al., 2006), cellular telephones (Jayaraman et al., 1999), computers (White et al., 2003), pulp and paper industry (Frota Neto et al., 2008) because they can achieve additional profit.

While the LCA principle has been successfully applied to design new products and processes that reduce environmental damage (global warming, ozone depletion, acidification, toxicity, etc.), limited work has been conducted on the development of decision making models that integrate both LCA principles and supply chain management principles (Seuring and Muller, 2008). In addition, few studies have addressed the impact of integrating external control mechanisms on sustainable supply chain management practices such as environmental regulations, take-back legislation, GHG emissions, and carbon taxes, emission trading (Stranlund, 2007) and carbon markets (Johnson and Heinen, 2004; Peace and Juliani, 2009). For instance, Nagurney et al. (2006) is one of the first studies that addresses carbon taxes in the electric power supply chains (Nagurney et al., 2006). Subramanian et al. (2008) propose an approach to integrate environmental consideration within a managerial decision making framework. A non-linear mathematical programming model is introduced that allows the incorporation of traditional operations planning considerations (capacity, production and inventory) with environmental considerations (design, production and end-of-life). Decisions on the number of carbon credits purchased and sold in different periods are added under the limitation of carbon emissions.

Ramudhin et al. (2010) are the first to propose a carbon market sensitive strategic planning model for sustainable supply chain network design. They show that considerations of internal and external control mechanisms are of great importance to decision makers when designing sustainable supply chains. This paper extends the model presented in Ramudhin et al. (2010) by consideration of the LCA methodology to establish successful sustainable supply chains over time. The capability of the model is illustrated by an example of strategic planning in the aluminum supply chain.

3. Problem statement and methodology

Among the different approaches available to assess the environmental impact of processes and organizations, the LCA method seems to be the most promising. It aggregates the results of different aspects of environmental studies including GHG emissions that are recognized as the most harmful elements to the environment and responsible for climate change. GHG emissions are calculated based on emission factors and converted to carbon dioxide equivalent quantity (CO_2e).

Many countries are implementing various mechanisms to reduce GHG emissions including incentives or mandatory targets to reduce carbon footprint. Carbon taxes and carbon markets (emissions trading) are recognized as the most cost-effective mechanisms (Labatt and White, 2007). The basic idea is to put a price tag on carbon emissions and create new investment opportunities to generate a fund for green technology development (Bayon et al., 2007; Labatt and White, 2007). There are already a number of active carbon markets for GHG emissions such as the European Union Emission Trading Scheme (or EU ETS) in Europe, the largest multi-national GHG emissions trading scheme in the world, the New Zealand Emissions Trading Scheme (NZ ETS) in New Zealand, the Chicago Climate Exchange in United State (Johnson and Heinen, 2004; Peace and Juliani, 2009), and more recently the Montreal Climate Exchange in Canada.

Measuring and assessing carbon emissions becomes then an important step that can be achieved by LCA techniques and software (Rice et al., 1997). However, compliance with the environmental regulation of carbon emissions in a cost-effective manner is challenging. Thus, supply chain network design model had been revised to include the additional cost due to GHG emissions at all levels of the supply chain and social variables affecting the quality of life of the community in which the supply chain operates.

As shown in Fig. 1, an LCA based approach is necessary in order to establish the link between the critical inputs (raw material, energy, human, used product, etc.) and the output (products, GHG emissions, waste) at each node of the network over its entire life cycle. Strategic planning of sustainable supply chains should

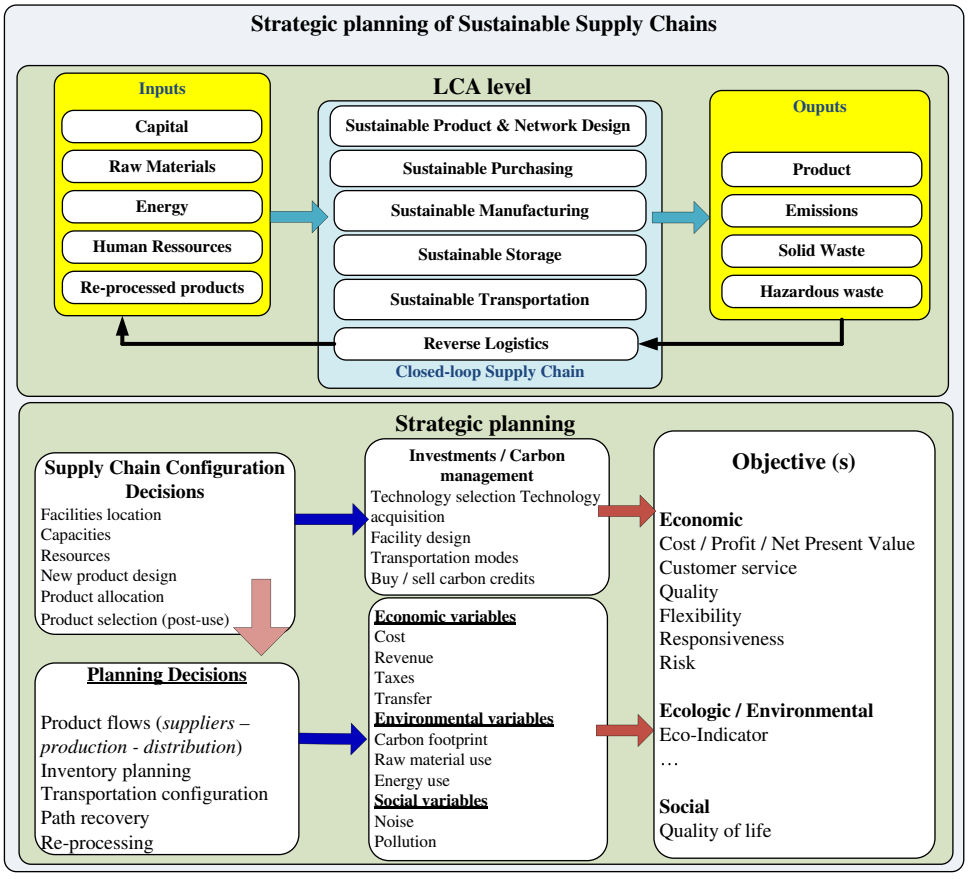


Fig. 1. An LCA approach to support sustainable supply chain design.

include the recovery of products decisions as well as carbon management strategies in order to be in compliance with the different environmental regulations. Thus, the supply chain performance should be evaluated based on the economic (cost and profit), the environmental (carbon emissions, recycling performance, waste management and energy use), and the social performances (quality of life, noise, etc.).

4. Model development

This section describes a generic mathematical model to help decision makers in the design and planning of sustainable supply chain based on the LCA methodology. The model establishes the link with the emission trading scheme to achieve sustainability objectives in a cost-effective manner under the different legislations that caps GHG emissions and impose mandatory targets for recycling products at the end of their life. Although supply chain sustainability recognizes the link between the economic, ecological, and social performance, an examination of social performances (labour equity, healthcare, safety, philanthropic commitment) shows that they are dependent on the context of operation of the supply chain (type of the industry), the government policies, and cultural norms. Thus, without loss of generality, we do not include the social performance in the mathematical formulation.

4.1. Assumptions

Fig. 2 shows the structure of the global supply chain. The sites are located in different zones $z \in Z$. A set of potential suppliers $n \in S$ can supply raw materials $p \in P^{MP}$ to a set of sub-contractors and plants $n \in F$ to manufacture products $p \in P^{PF}$. The latter can be distributed through a set of potential distribution centers $n \in D$. Final products are shipped from the distribution centers to different customers or markets $n \in C$. Also available are different recycling centers $n \in R$ for the processing of used products that can be returned to different stages in the supply chain. Let N denotes the set of the different nodes of the supply chain network, $N = S \cup F \cup D \cup C \cup R$.

At each production center, a set of potential technologies $h \in H$ is available for use. Each of these technologies needs some inputs (energy, liquid, solid, gazes, etc.) $i \in I$, in addition to materials and generate different outputs (liquid, solid, gazes) $o \in O$. Different transportation modes $m \in M$ are used for the shipment of products between nodes (suppliers, production units, distribution centers, and recycling units). Each transportation mode needs some inputs (e.g. energy and gazes) and may generate some wastes (output). The main objective of the model is to support sustainable supply chain network design over a long-term period of time $t \in T$.

Before the description of the detailed model, some basic elements about modeling techniques are explained. Generally, we define two types of nodes (production unit and recycling center). For a production unit, one or several technologies will be available for manufacturing activities. The production is based on a bill of material that indicates the quantity of raw material or components required to manufacture components or final products. The potential technologies available differ in terms of acquisition and operation costs as well as inputs consumption and output emissions. Fig. 3 summarize the situation.

For a recycling center (Fig. 4), we consider that a certain percentage of products are recovered by the company. To simplify the calculation, we consider that is proportional to the demand at each period. Thus, we assume that for each period, there is a quantity of used products collected and delivered to recycling centers. Using a bill of material of disassembly, the product is disassembled. Good

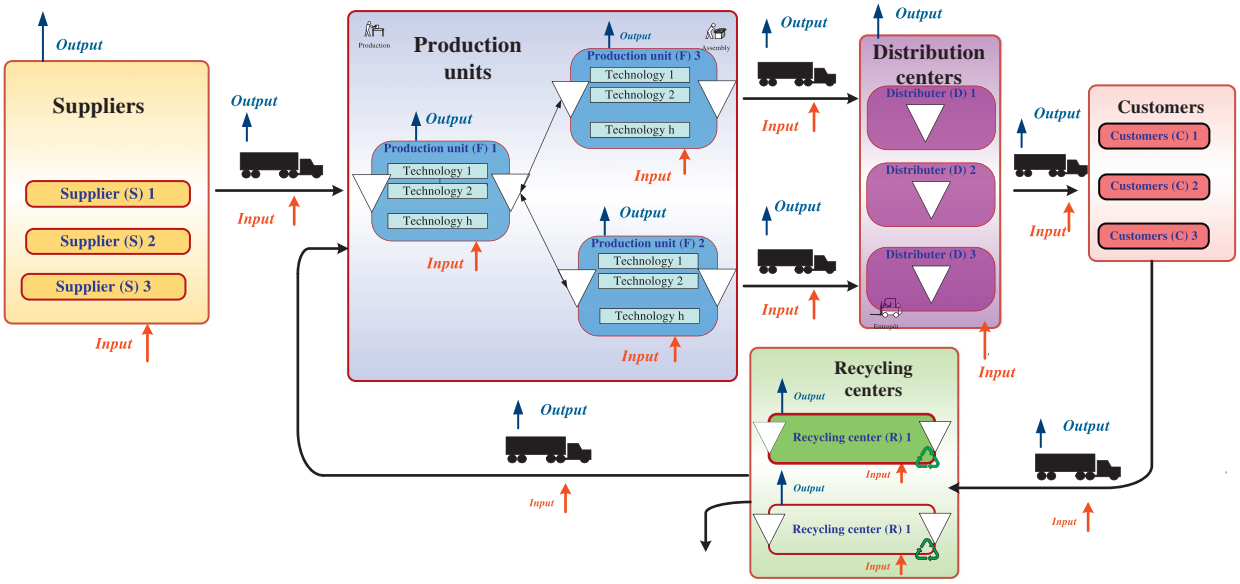


Fig. 2. Closed-loop supply chain network structure.

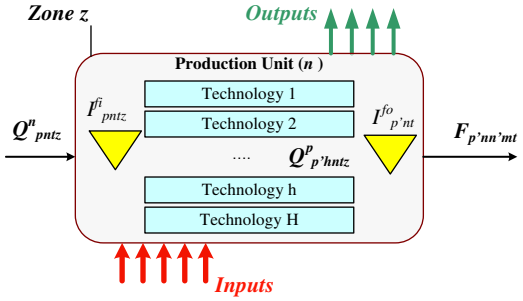


Fig. 3. Characteristics of a production unit.

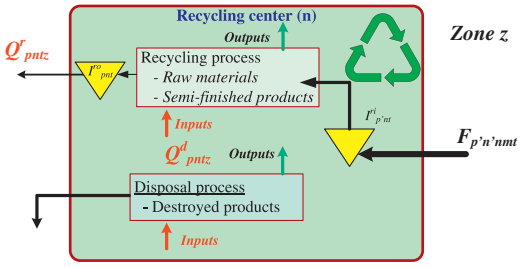


Fig. 4. Characteristics of a recycling center.

recycled components and raw materials are reintegrated in production units. The non-useable products (components and materials) are destroyed thorough the disposal process.

4.2. Decision variables

To achieve the objective of sustainability and understand the impact of different control parameters on the decision process, several decisions touching various aspects of the supply chain must be taken into account. They are:

a) Decisions related to sites (plants and recycling centers) location

Y^l_{nz} Binary variable takes a value of 1 if site $n \in F \cup D \cup R$ is located at zone $z \in Z$, 0 otherwise.

b) Decisions related to production units

Q^n_{pnt} Quantity of product $p \in P^{MP} \cup P^{PF}$ necessary in production unit $n \in F$ during time period $t \in T$.

Q^p_{phntz} Quantity of product $p \in P^{PF}$ manufactured/assembled using technology $h \in H$ at production unit $n \in F$ during time period $t \in T$ at zone $z \in Z$.

I^{fi}_{pnt} The inventory level of input product $p \in P^{MP} \cup P^{PF}$ at unit production $n \in F$ during time period $t \in T$.

I^{fo}_{pnt} The inventory level of output product $p \in P^{PF}$ at unit production $n \in F$ during time period $t \in T$.

Y^s_{pnt} Binary variable takes a value of 1 if supplier $n \in S$ is selected for supplying raw material $p \in P^{MP}$ during time period $t \in T$, 0 otherwise.

Y^f_{nt} Binary variable takes a value of 1 if production unit $n \in F$ is operational during time period $t \in T$, 0 otherwise.

Y^h_{hnt} Binary variable takes a value of 1 if technology $h \in H$ is selected at production unit $n \in F$ during time period $t \in T$, 0 otherwise.

Y^p_{phnt} Binary variable takes a value of 1 if product $p \in P^{PF}$ is manufactured/assembled using technology $h \in H$ at production unit $n \in F$ during time period $t \in T$, 0 otherwise.

c) Decisions related to distribution centers

Y^d_{nt} Binary variable takes a value of 1 if distribution center $n \in D$ is operational during time period $t \in T$, 0 otherwise.

I^d_{pnt} The inventory level of product $p \in P^{PF}$ at distribution center $n \in D$ during time period $t \in T$.

d) Decisions related to recycling centers

Y^r_{nt} Binary variable takes a value of 1 if recycling center $n \in R$ is operational during time period $t \in T$, 0 otherwise.

Q^a_{pntz} Quantity of product $p \in P^{PF}$ refurbished at recycling center $n \in R$ during time period $t \in T$ at zone $z \in Z$.

Q^r_{pntz} Quantity of raw material $p \in P^{MP}$ recycled at recycling center $n \in R$ during time period $t \in T$ at zone $z \in Z$.

Q^d_{pntz} Quantity of product $p \in P^{MP} \cup P^{PF}$ disposed at recycling center $n \in R$ during time period $t \in T$ at zone $z \in Z$.

I^{ri}_{pnt} The inventory level of input of product $p \in P^{PF}$ at recycling center $n \in R$ during time period $t \in T$.

I^{ro}_{pnt} The inventory level of output of recycled material $p \in P^{MP} \cup P^{PF}$ at recycling center $n \in R$ during time period $t \in T$.

e) Decisions related to transportation

$F_{pnn'mt}$ Quantity of product $p \in P^{MP} \cup P^{PF}$ processed from node $n \in N$ to node $n' \in N$ using transportation mode $m \in M$ during time period $t \in T$.

$Y^{t}_{mnn't}$ Binary variable takes a value of 1 if transportation mode $m \in M$ is selected between node $n \in N$ and node $n' \in N$ during time period $t \in T$, 0 otherwise.

f) Decisions related to carbon management

CC^{+}_{tz} Credits purchased during time period $t \in T$ at zone $z \in Z$.

CC^{-}_{tz} Credits sold during time period $t \in T$ at zone $z \in Z$.

Decisions on (a), (b), (c), (d) and (e) are related to the supply chain network configuration and aggregate planning, while decisions on (f) determine the strategy to put in place for GHG emissions control. The problem can be viewed as a multi-objective model where the economic objective function, denoted by $\mathbf{F_1}$, evaluates the total logistic cost and the environmental objective function, denoted by $\mathbf{F_2}$, evaluates carbon emissions resulting from operation strategies, manufacturing, and transportation activities. Using the Intergovernmental Panel for Climate Change (IPCC) guidelines and other environmental data, the global warming potential (GWP) for each activity of the supply chain is expressed in terms of carbon dioxide equivalent quantity (CO_2e). Emissions are assumed to be linearly proportional to manufacturing, transportation, and usage.

5. Model formulation

This section describes the linear programming model that considers the critical aspects for the design and strategic planning of sustainable supply chains. The choice of multi-objective linear programming (MOLP) as a methodology to investigate this problem is basically because it helps to find the different strategic decisions (explained in the previous framework: see Fig. 1) of linear objective functions and a single decision maker or a decision making body (the focal company) that guarantee a trade-off with respect to some linear constrains.

5.1. Economic objective ($\boldsymbol{F_1}$)

The strategic sustainable supply chain network design described before has the objective to find a trade-off solution between the economic and the environmental performance under the different regulations that caps GHG emissions and impose constraints related producer responsibility at the end of the production life cycle. The economic objective is evaluated by the total logistic cost. The environmental performance is evaluated by the total emissions of GHG.

The economic dimension includes different costs

- *Location cost* (*denoted ZC*): which are the costs to locate production, distribution, and recycling centers at the different regions

$$\sum_{n \in F \cup D \cup R} \sum_{z \in Z} a^{l}_{nz} Y^{l}_{nz}$$

- *Supply costs* (*denoted SC*): which are the costs to acquire materials

  Fixed cost to establish contracts with suppliers

$$\sum_{p \in P^{MP}} \sum_{n \in S} \sum_{t \in T} a^{s}_{pnt} Y^{s}_{pnt}$$

  Variable cost for raw materials acquisition

$$\sum_{p \in P^{MP}} \sum_{n \in S} \sum_{n' \in F} \sum_{m \in M} \sum_{t \in T} c^{af}_{pnt} F_{pnn'mt}$$

  Recycled materials acquisition

$$\sum_{p \in P^{MP}} \sum_{n \in R} \sum_{n' \in F} \sum_{m \in M} \sum_{t \in T} c^{ar}_{pnt} F_{pnn'mt}$$

- *Production costs* (*denoted PC*): which are the costs to manufacture products

  Fixed cost for operating production units

$$\sum_{n \in F} \sum_{t \in T} a^{f}_{nt} Y^{f}_{nt}$$

  Fixed cost for technology acquisition

$$\sum_{h \in H} \sum_{n \in F} \sum_{t \in T} a^{h}_{hnt} Y^{h}_{hnt}$$

  Fixed cost for production line configuration

$$\sum_{p \in P^{PF}} \sum_{h \in H} \sum_{n \in F} \sum_{t \in T} a^{p}_{phnt} Y^{p}_{phnt}$$

  Variable cost for manufacturing

$$\sum_{p \in P^{PF}} \sum_{h \in H} \sum_{n \in F} \sum_{t \in T} \sum_{z \in Z} c^{p}_{phntz} Q^{p}_{phntz}$$

  Inventory cost of materials

$$\sum_{p \in P^{MP}} \sum_{n \in F} \sum_{t \in T} c^{fi}_{pnt} I^{fi}_{pnt}$$

  Inventory cost of products

$$\sum_{p \in P^{PF}} \sum_{n \in F} \sum_{t \in T} c^{fo}_{pnt} I^{fo}_{pnt}$$

- *Distribution costs* (*denoted DC*): which are the costs to distribute products

  Fixed cost for operating distribution centers

$$\sum_{n \in D} \sum_{t \in T} a^{d}_{nt} Y^{d}_{nt}$$

  Variable cost for material handling products

$$\sum_{p \in P^{PF}} \sum_{n \in D} \sum_{t \in T} c^{d}_{pnt} I^{d}_{pnt}$$

- *Reverse logistics costs* (*denoted RC*): which are the costs to recycle and dispose products

  Fixed cost for operating recycling centers

$$\sum_{n \in R} \sum_{t \in T} a^{r}_{nt} Y^{r}_{nt}$$

  Cost of recovery of used products

$$\sum_{p \in P^{PR}} \sum_{n \in R} \sum_{n' \in C} \sum_{m \in M} \sum_{t \in T} c^{rc}_{pnt} F_{pn'nmt}$$

  Variable cost of recycling used products

$$\sum_{p \in P^{PF}} \sum_{n \in R} \sum_{t \in T} \sum_{z \in Z} c^{r}_{pntz} Q^{r}_{pntz}$$

  Variable cost for disposal of used products

$$\sum_{p \in P^{PF}} \sum_{n \in R} \sum_{t \in T} \sum_{z \in Z} c^{dr}_{pntz} Q^{d}_{pntz}$$

  Inventory cost for recovered used products

$$\sum_{p \in P^{PF}} \sum_{n \in R} \sum_{t \in T} c^{ri}_{pnt} I^{ri}_{pnt}$$

  Inventory cost for recycled products

$$\sum_{p \in P^{MP} \cup P^{PF}} \sum_{n \in R} \sum_{t \in T} c^{ro}_{pnt} I^{ro}_{pnt}$$

- *Transportation cost* (*denoted TC*): which are the costs to move products

  Fixed cost for transportation links between nodes

  $$\sum_{n\in N}\sum_{n'\in N}\sum_{m\in M}\sum_{t\in T} a^t_{nn'mt}Y^t_{nn'mt}$$

  Variable cost for transportation

  $$\sum_{p\in P^{MP}\cup P^{PF}}\sum_{n\in N}\sum_{n'\in N}\sum_{m\in M}\sum_{t\in T} c^t_{pnn'mt}F_{pnn'mt}$$

- *LCA based cost* (*denoted LC*): We consider that the company will identify some strategic input costs (water, oil, energy, etc.) that need to be considered in economic objective function. Also, some outputs (waste, co-products, etc.) need further treatment and there are also some related costs. Let us denote C_{it} the consumption of the input $i\in I$ during period $t\in T$

$$C_{it}=\left(\sum_{p\in P^{PF}}\sum_{h\in H}\sum_{n\in F}\sum_{z\in Z} CF^p_{iph}Q^p_{phntz}\right)+\left(\sum_{p\in P^{PF}}\sum_{n\in R}\sum_{z\in Z} CF^r_{ipn}Q^r_{pntz}\right)+\left(\sum_{p\in P^{PF}}\sum_{n\in R}\sum_{z\in Z} CF^d_{ipn}Q^d_{pntz}\right)+\left(\sum_{p\in P}\sum_{n\in N}\sum_{n'\in N}\sum_{m\in M} CF^t_{im}F_{pnn'mt}\right)\quad \forall i\in I,\ \forall t\in T \tag{1}$$

Let us denote E_{ot} the emission of the output $o\in O$ during period $t\in T$

$$E_{ot}=\left(\sum_{p\in P^{PF}}\sum_{h\in H}\sum_{n\in F}\sum_{z\in Z} EF^p_{oph}Q^p_{phntz}\right)+\left(\sum_{p\in P^{PF}}\sum_{n\in R}\sum_{z\in Z} EF^r_{opn}Q^r_{pntz}\right)+\left(\sum_{p\in P^{PF}}\sum_{n\in R}\sum_{z\in Z} EF^d_{opn}Q^d_{pntz}\right)+\left(\sum_{p\in P}\sum_{n\in N}\sum_{n'\in N}\sum_{m\in M} EF^t_{om}F_{pnn'mt}\right)\quad \forall o\in O,\ \forall t\in T \tag{2}$$

Thus, the cost of using inputs and treating outputs (if necessary) is

$$\sum_{i\in I}\sum_{t\in T} v_{it}C_{it}+\sum_{o\in O}\sum_{t\in T} u_{ot}E_{ot}$$

- *Carbon credit component* (*denoted CC*): For many organizations and industrial sectors, the main emissions are greenhouse gases. Many companies have set voluntary targets in term of GHG emissions attributable to their supply chain or are subject to a new regulation that "caps" GHG emissions. Under an Emission trading Scheme (ETS), carbon dioxide (CO_2) is tradable. This system is based on the allocation of units to a company for exceeding its intensity-based GHG emissions reduction targets [1 credit=right to emit one metric ton of carbon dioxide equivalent (CO_2e)]. At the end of each compliance period, the emissions of the company will be verified. Each emitter must then offset its GHG emissions against its intensity-based GHG emissions reduction target established by the government. The discrepancy between the imposed target and the actual emissions may be offset by, among other things, the purchase of units on the domestic market. In addition to internal reductions, large emitters will be able to buy units from the carbon market in order to ensure compliance with their GHG emissions reductions obligations. On the other hand, those companies with emissions less than the cap will have the possibility to sell credits in the carbon market and generate profit. Thus, "carbon management" consists of taking the decision on the most cost-effective strategy to be in compliance either with environmental regulation or with voluntary targets. Thus, the decision is to determine the number of credits purchased (A^{CC}_{tz}) in period $t\in T$ and the number of credits sold (V^{CC}_{tz}) in period $t\in T$.

$$CC=\sum_{t\in T}\sum_{z\in Z} CC^-_{tz}A^{CC}_{tz}-\sum_{t\in T}\sum_{z\in Z} CC^+_{tz}V^{CC}_{tz} \tag{3}$$

In summary, the economic performance in Eq. (4) is measured by the objective function $\mathbf{F_1}$ that should be minimized to ensure economic sustainability.

$$\mathbf{F_1}=ZC+SC+PC+DC+RC+TC+LC+CC \tag{4}$$

5.2. Environmental objective ($\mathbf{F}_2$)

The second key objective to achieve sustainable supply chains is the evaluation and the optimization of the environmental impact (Eq. (5)). The determination of the environmental performance of a supply chain is not easy and might be different from one industry sector to another. However, the use of an LCA approach helps in the evaluation of the environmental performance of product, process, and service. To make it general, we aggregate the different impacts in term of GHG emissions (objective function $\mathbf{F_2}$) which are very important in our case (due to the link with ETS). Once again, GHG emissions should be minimized to ensure environmental sustainability.

$$\mathbf{F_2}=\sum_{t\in T}\left(\sum_{o\in O}E_{ot}CO^{out}_o+\sum_{i\in I}C_{it}CO^{in}_i\right) \tag{5}$$

5.3. Constraints

5.3.1. Suppliers

Supplier's capacity

$$\sum_{n'\in F}\sum_{m\in M} F_{pnn'mt}\le\lambda_{pnt}Y^s_{pnt}\quad \forall p\in P^{MP},\ \forall n\in S,\ \forall t\in T \tag{6}$$

If the supplier is selected, it will stay operational for the whole planning horizon

$$Y^s_{pnt}\ge Y^s_{pn(t-1)}\quad \forall p\in P^{MP},\ \forall n\in S,\ \forall t\in T \tag{7}$$

5.3.2. Production units

Location of production units at zones

$$\sum_{p\in P^{PF}}\sum_{h\in H}\sum_{t\in T} Q^p_{phntz}\le MY^l_{nz},\quad \forall n\in F,\ \forall z\in Z,\ M \text{ big number} \tag{8}$$

Raw material products usage

$$Q^n_{pnt}=\sum_{p'\in P^{PF}}\sum_{h\in H}\sum_{z\in Z}\varphi_{pp'}Q^p_{p'hntz}\quad \forall p\in P^{MP},\ \forall n\in F,\ \forall t\in T \tag{9}$$

Products usage

$$\sum_{n'\in F\cup R}\sum_{m\in M} F_{pn'nmt}=\sum_{p'\in P^{PF}}\sum_{h\in H}\sum_{z\in Z}\varphi_{pp'}Q^p_{p'hntz}\quad \forall p\in P^{PF},\ \forall n\in F,\ \forall t\in T \tag{10}$$

Capacity of production units

$$\sum_{z\in Z} Q^p_{phntz}\le q^p_{phnt}Y^p_{phnt}\quad \forall p\in P^{PF},\ \forall h\in H,\ \forall n\in F,\ \forall t\in T \tag{11}$$

Logic constraints: if a technology is not selected at a production unit, there is no need for configuration

$$Y^p_{phnt}\le Y^h_{hnt}\quad \forall p\in P^{PF},\ \forall n\in F,\ \forall h\in H,\ \forall t\in T \tag{12}$$

Logic constraints: if the production unit is not operational, there is no need to implement a technology in this facility

$$Y^h_{hnt} \le Y^f_{nt} \quad \forall n \in F, \forall h \in H, \forall t \in T \tag{13}$$

Inventory of materials at production units

$$I^{fi}_{pn(t-1)} + \sum_{n' \in S} \sum_{m \in M} F_{pn'nmt} + \sum_{n' \in R} \sum_{m \in M} F_{pn'nmt} = I^{fi}_{pnt} + Q^n_{pnt} \quad \forall p \in P^{MP}, \forall n \in F, \forall t \in T \tag{14}$$

Initial inventory levels for products

$$I^{fi}_{pn0} = 0 \quad \forall p \in P^{MP} \cup P^{PF}, \forall n \in F \tag{15}$$

Inventory capacity constraints (raw material, components)

$$I^{fi}_{pnt} \le i^{fi}_{pnt} \quad \forall p \in P^{MP} \cup P^{PF}, \forall n \in F, \forall t \in T \tag{16}$$

Inventory of output products

$$I^{fo}_{pn(t-1)} + \sum_{h \in H} \sum_{z \in Z} Q^p_{phntz} + \sum_{n' \in R} \sum_{m \in M} F_{pn'nmt} = I^{fo}_{pnt} + \sum_{n' \in F \cup D} \sum_{m \in M} F_{pnn'mt} \quad \forall p \in P^{PF}, \forall n \in F, \forall t \in T \tag{17}$$

Initial inventory levels for products

$$I^{fo}_{pn0} = 0 \quad \forall p \in P^{PF}, \forall n \in F \tag{18}$$

Inventory capacity constraints (components, products)

$$I^{fo}_{pnt} \le i^{fo}_{pnt} \quad \forall p \in P^{PF}, \forall n \in F, \forall t \in T \tag{19}$$

If a production unit is operational, it will stay for the whole planning horizon

$$Y^f_{nt} \ge Y^f_{n(t-1)} \quad \forall n \in F, \forall t \in T \tag{20}$$

If a technology is acquired, it is used for the whole horizon

$$Y^h_{hnt} \ge Y^h_{hn(t-1)} \quad \forall h \in H, \forall n \in F, \forall t \in T \tag{21}$$

Logic constraint for operating production units

$$Y^f_{nt} <= \sum_{z \in Z} Y^l_{nz} \quad \forall n \in F, \forall t \in T \tag{22}$$

Limited number of production units per zone

$$\sum_{z \in Z} Y^l_{nz} <= 1 \quad \forall n \in F \tag{23}$$

5.3.3. Distribution centers (DCs)

Inventory constraints at distribution centers

$$I^d_{pn(t-1)} + \sum_{n' \in F} \sum_{m \in M} F_{pn'nmt} = I^d_{pnt} + \sum_{n' \in C} \sum_{m \in M} F_{pnn'mt} \quad \forall p \in P^{PF}, \forall n \in D, \forall t \in T \tag{24}$$

Initial inventory levels for final products

$$I^d_{pn0} = 0 \quad \forall p \in P^{PF}, \forall n \in D \tag{25}$$

Inventory capacity constraints for final products at DCs

$$I^d_{pnt} \le i^d_{pnt} \quad \forall p \in P^{PF}, \forall n \in D, \forall t \in T \tag{26}$$

Distribution center capacity

$$\sum_{n' \in F} \sum_{m \in M} F_{pn'nmt} + \sum_{n' \in C} \sum_{m \in M} F_{pnn'mt} \le \chi_{nt} Y^d_{nt} \quad \forall p \in P^{PF}, \forall n \in D, \forall t \in T \tag{27}$$

If the production center is selected, it will stay operational for the whole planning horizon

$$Y^d_{nt} \ge Y^d_{n(t-1)} \quad \forall n \in D, \forall t \in T \tag{28}$$

If the distribution center is not located in a specific region then it is not operational

$$Y^d_{nt} \le \sum_{z \in Z} Y^l_{nz} \quad \forall n \in D, \forall t \in T \tag{29}$$

Limited number of distribution centers per zone

$$\sum_{z \in Z} Y^l_{nz} <= 1 \quad \forall n \in D \tag{30}$$

5.3.4. Customers

Demand constraint

$$\sum_{n' \in D} \sum_{m \in M} F_{pn'nmt} = d_{pnt} \quad \forall p \in P^{PF}, \forall n \in C, \forall t \in T \tag{31}$$

5.3.5. Recycling centers

Location recycling centers at zones

$$\sum_{p \in P^{PF}} \sum_{z \in Z} (Q^r_{pntz} + Q^d_{pntz} + Q^a_{pntz}) \le M Y^l_{nt}, \quad \forall n \in R, \forall t \in T, M \text{ big number} \tag{32}$$

Recovery of used products

$$\sum_{n \in C} \sum_{n' \in R} \sum_{m \in M} F_{pnn'mt} = \sum_{n \in N} \delta_{pt} d_{pnt} \quad \forall p \in P^{PF}, \forall t \in T \tag{33}$$

Initial inventory of products recovered at recycling centers

$$I^{ri}_{pn0} = 0 \quad \forall p \in P^{PF}, \forall n \in R \tag{34}$$

Inventory of used products at recycling centers

$$I^{ri}_{pn(t-1)} + \sum_{n' \in C} \sum_{m \in M} F_{pnn'mt} = I^{ri}_{pnt} + \sum_{z \in Z} Q^a_{pntz} \quad \forall p \in P^{PF}, \forall n \in R, \forall t \in T \tag{35}$$

Inventory capacity of used products of recycling centers

$$I^{ri}_{pnt} \le i^{ri}_{pnt} \quad \forall p \in P^{PF}, \forall n \in R, \forall t \in T \tag{36}$$

Disposal of non valuable products (sorting process)

$$Q^d_{pntz} = \theta_{pntz} Q^a_{pntz} \quad \forall p \in P^{PF}, \forall n \in R, \forall t \in T, \forall z \in Z \tag{37}$$

Re-processing of good products

$$Q^r_{pntz} = \sum_{p' \in P^{PF}} \phi_{pp'} (1 - \theta_{p'ntz}) Q^a_{p'ntz} \quad \forall p \in P^{MP} \cup P^{PF}, \forall n \in R, \forall t \in T, \forall z \in Z \tag{38}$$

Inventory of output products (raw material, components) from recycling centers

$$I^{ro}_{pn(t-1)} + \sum_{z \in Z} Q^r_{pntz} = I^{ro}_{pnt} + \sum_{n' \in F} \sum_{m \in M} F_{pnn'mt} \quad \forall p \in P^{MP} \cup P^{PF}, \forall n \in R, \forall t \in T \tag{39}$$

Initial inventory level of output products (raw material and components) from recycling centers

$$I^{ro}_{pn0} = 0 \quad \forall p \in P^{MP} \cup P^{PF}, \forall n \in R \tag{40}$$

Inventory capacity of output products (raw material and components) at recycling centers

$$I^{ro}_{pnt} \le i^{ro}_{pnt} \quad \forall p \in P^{MP} \cup P^{PF}, \forall n \in R, \forall t \in T \tag{41}$$

Recycling process capacity

$$\sum_{z \in Z} Q^a_{pntz} \le q^r_{pnt} Y^r_{nt} \quad \forall p \in P^{PF}, \forall n \in R, \forall t \in T \tag{42}$$

If a node is operational, it is used for the planning horizon

$$Y^r_{nt} \ge Y^r_{n(t-1)} \quad \forall n \in D, \forall t \in T \tag{43}$$

5.3.6. Transportation

Transportation capacity

$$\sum_{p \in P^{MP} \cup P^{PF}} F_{pnn't} \le c_{mnn't} Y^t_{mnn't} \quad \forall n \in N, \forall n' \in N, \forall m \in M, \forall t \in T \tag{44}$$

If the recycling center is not located in a specific region then it is not operational

$$Y_{nt}^{R} \leq \sum_{z \in Z} Y_{nz}^{l} \quad \forall n \in R, \forall t \in T \tag{45}$$

Limited number of distribution centers per zone

$$\sum_{z \in Z} Y_{nz}^{l} <= 1 \quad \forall n \in D \tag{46}$$

5.3.7. Carbon management

At each period $\forall t \in T$ and zone $\forall z \in Z$, the company must be in compliance with the limitation of carbon emissions (CO_2e), thus

$$\sum_{o \in O} CO_{o}^{out} \left[\sum_{p \in P^{PF}} \sum_{h \in H} \sum_{n \in F} EF_{oph}^{p} Q_{phntz}^{p} + \sum_{p \in P^{PF}} \sum_{n \in R} EF_{opn}^{r} Q_{pntz}^{r} + \sum_{p \in P^{PF}} \sum_{n \in R} EF_{opn}^{d} Q_{pntz}^{d} \right] + \sum_{i \in I} CO_{i}^{in} \left[\sum_{p \in P^{PF}} \sum_{h \in H} \sum_{n \in F} CF_{iph}^{p} Q_{phntz}^{p} + \sum_{p \in P^{PF}} \sum_{n \in R} CF_{ipn}^{r} Q_{pntz}^{r}) + \sum_{p \in P^{PF}} \sum_{n \in R} CF_{ipn}^{d} Q_{pntz}^{d} \right] + CC_{tz}^{+} - CC_{tz}^{-} \leq L_{tz}^{CO_2 e} \quad \forall t \in T \forall z \in Z \tag{47}$$

Limit on the number of credits to purchase

$$CC_{tz}^{-} \leq L_{tz}^{COP} \quad \forall t \in T \forall z \in Z \tag{48}$$

Limit on the number of credits that can be sold

$$CC_{tz}^{+} \leq L_{tz}^{COS} \quad \forall t \in T \forall z \in Z \tag{49}$$

6. Experimental evaluation

6.1. Data

The mathematical model has been developed, validated and was used in a preliminary study of a supply chain from the aluminum industry to illustrate the potential application as a decision making tool for sustainable supply chain planning under different environmental regulations that impose mandatory limits on carbon emissions as well as the obligation of product recovery and recycling at the end of life.

In this research, we consider the aluminum production as an example to study some important research questions and find some managerial decisions under environmental regulations. In the aluminum industry, there are typically two sources of raw materials, namely, bauxite which is the primary raw material from which aluminum is made, and secondary aluminum which is obtained by recycling aluminum products. Since aluminum is 100% recyclable without any loss of its natural qualities, recovery of the metal via recycling has become an important facet of the industry. Products under consideration can be made either made from primary or secondary aluminum using either one of two potential technologies which have different operating costs and different GHG emissions. Critical inputs and outputs including liquid, solid, energy, and gaseous wastes are considered.

We consider that the supply chain have two type of family products. Two production units are responsible for the production and products are distributed using two distribution centers. For each raw material, two suppliers are available. Used products are returned to recycling centers. Depending on the state of the returned product, they are recycled or disposed. The recycled products are returned to the production units to be used in the production process. An overview of the supply chain is shown in Fig. 5. Some statistics for the model are summarized in Table 1.

It is clear from the model formulation in the previous section that some data to use in the model is sometimes difficult to find and the company should make an effort to collect a big number of input parameters to use the proposed model. However, the primary goal in this paper is to demonstrate how the mathematical model can be used in order to evaluate the impact of different legislations on the supply chain strategy and planning. In practice, the model could be modified to capture some specific strategic considerations of a given supply chain and populated with additional data.

For the aim of the study, some parameters are estimated, the others ones are collected from the available information on the web. For example, the model developed in this paper highlights the importance of LCA data in order to help and inform managerial decision making. Numeric estimates of air emissions, solid waste generation, material consumption, are valuable in estimating model's parameters, thus enabling the bridging of operational and environmental decisions. The Intergovernmental Panel on Climate Change (IPCC), is an important source of data and was used to estimate emission factors (http://www.ipcc-nggip.iges.or.jp/EFDB/main.php).

6.2. Solution method

Different commercially-available optimization software exists today to solve MIP optimization problems. As discussed in previous section, it is currently impossible to use real data to populate the model. However, to demonstrate the practical solvability of the model, we randomly generate some numerical scenarios of parameters and constraints (see Table 2), and solve the model using the LINGO version of LINDO systems Inc on an AMD 2 493 Mhz PC running Windows XP.

The mean and median run times per numerical scenario with the default LINGO solver settings, were 600 and 500 s, respectively, which is a very practical amount of time. Although this example is with limited number of products and sites (suppliers, production units, and recycling centers), more sophisticated global optimization approaches for large scale optimization problem could be used to solve large scale supply chain network.

6.3. Research questions

In this section, we will present briefly the research questions that we attempt to answer by populating the model with data and solving it. To be in compliance with the environmental regulations, the company has different options. The first one is to re-locate production units and recycling centers at other regions. Carbon prices vary from one region to another. The company can invest on new production technologies to reduce carbon emissions and energy use. Thus the first research question is: Given the regulation on carbon emissions, what is the nature of compliance of the supply chain given the cost of the various compliance options (the cost of green technologies versus carbon emission trading)? The second research question is: Given the regulation based on take back legislation that impose collection, recovery and recycling targets for products at the end of their life, what is the nature of managerial decisions related to the supply chain?

6.4. Results and discussion

The first aspect analyzed is the impact of carbon price variations on the supply chain configuration under two different scenarios. In scenario 1, carbon prices are stable in time. However, in scenario 2, carbon prices increase over time. The carbon prices (Fig. 6) and results for scenario 2 are shown in Fig. 7. Here, the carbon credit component is positive and represents 7% of the total cost. That

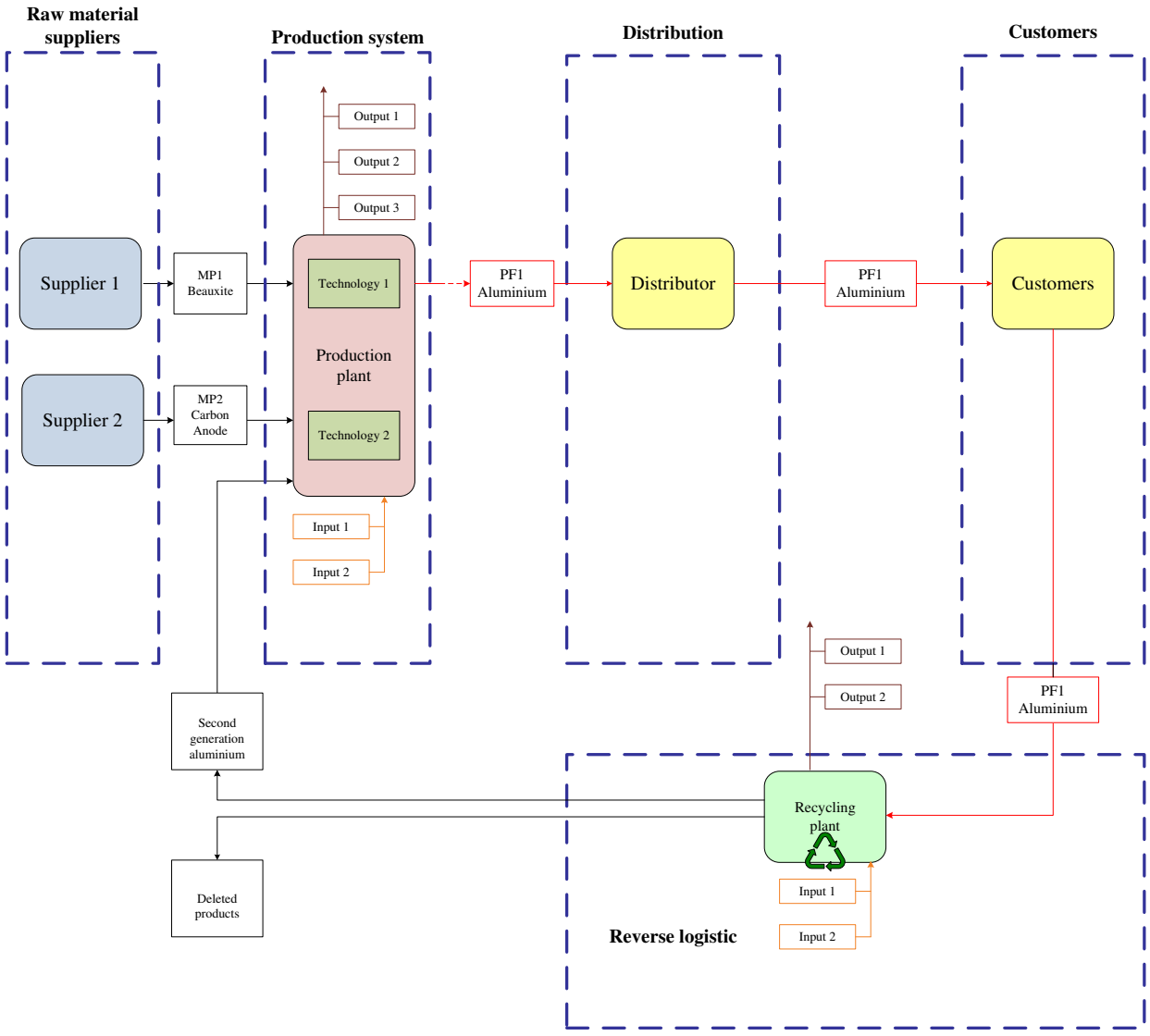


Fig. 5. Case study supply chain network.

Table 1
Statistics for the model.

Model components	Quantity	Model components	Quantity
Periods	4	Production technologies	2
Materials	2	Distribution centers	2
Products	2	Clients	2
Regions	2	Recycling unit	2
Suppliers	2	Inputs	2
Production units	2	Outputs	2

Table 2
Characteristics of the MIP model.

	Number of variables	Binary variables	Continuous variables	Number of constraints
Case study	1136	292	844	809

means that the supply chain needs to buy $1 441 320 worth of carbon credits during the planning period to be in compliance with the environmental regulation.

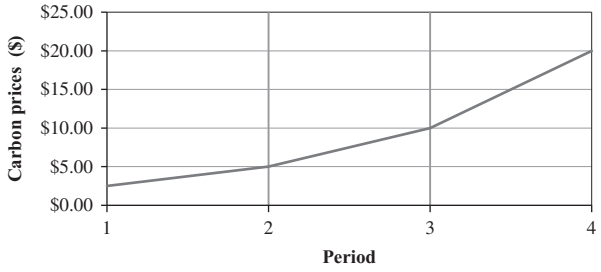


Fig. 6. Carbon prices variation for scenario 2.

Table 3 compares the results obtained for the two scenarios. First, we observe that the emission cost for scenario 2 is higher but that the total logistic cost remains the same for both scenarios. This is because the supply chain configuration (combination of sites, technology used, distribution channels, etc.) is the same in both scenarios. Here carbon prices only resulted in an increase in total cost with no consequence on the supply chain configuration because the marginal cost for reducing one unit of GHG emissions is greater than the carbon price from the market. Hence, the best

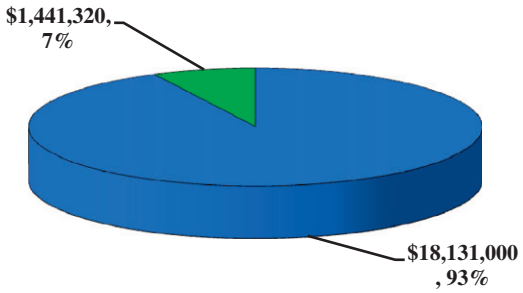


Fig. 7. Cost distribution for scenario 2.

Table 3
Comparison of the two scenarios.

	Scenario 1 Stable	**Scenario 2** Increase
Total logistics cost	$18 131 000	$18 131 000
Cost of carbon credit	$1 216 320	$1 441 320
Total cost	$19 347 300	$19 572 300

Table 4
Cost for the different scenarios (Return rate variation).

	Scenario 1 (δ=80%)	**Scenario 2 (δ=120%)**
Total cost	$19 347 300	$20 929 800
Total logistics cost	$18 131 000	$19 214 100
Carbon credit	$1 216 320	$1 715 700

decision is to buy credits form the carbon market to be in compliance with the regulation limits on carbon emissions.

The second aspect analyzed is the impact of recycling strategies on supply chain planning decisions. Here we assume that legislation forces the company to accept all recycled products first. The supply chain is solved for different return rates (δ) of aluminum products. For the first scenario, we consider that only 80% of products available in the market are recycled (δ=80%). In the second scenario, secondary aluminum may come from other sources including the direct customers and hence a return rate of 120% (δ=120%). Table 4 summarizes the results obtained in this case. It shows that an increase in recycling of the products increases the total cost by 8.2% which translates into a 5.9% increase in logistics cost and a 41.1% increase in carbon credit cost. In this case, the legislation on recycling has a negative impact on carbon costs as it forces the supply chain to use technologies that have higher GHG emissions.

The final aspect studied is the impact of limit on emissions (LCO_2). We analyze two scenarios where regulations in terms of carbon emissions becomes more stringent (LCO_2=25 000 tCO_2e versus LCO_2=5000 tCO_2e). In this case, we suppose that carbon prices will increase (Fig. 6). Fig. 8 shows that the quantity of recycled product increases as the limit of emissions is more stringent because carbon emissions are reduced due to the use of recycled product. When recycling is cheaper and with less GHG emissions, product recycling mostly increases and the cost is minimized.

However, in the last period, the quantity of recycled product decreases. Indeed, due to the strategy of carbon management (Fig. 9) that consists of buying carbon credits when carbon prices are not expensive helps the company to reduce the cost of

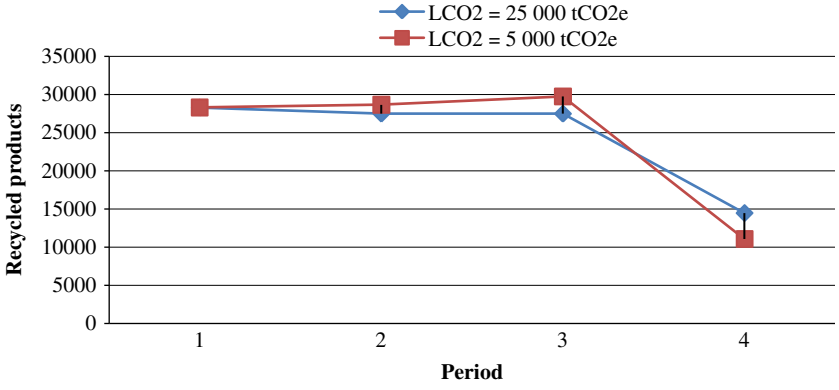


Fig. 8. Recycled product under policy stringency.

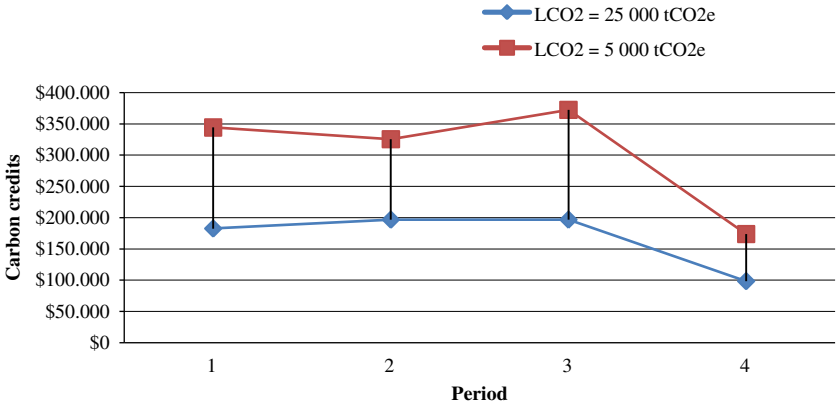


Fig. 9. Carbon management under policy stringency.

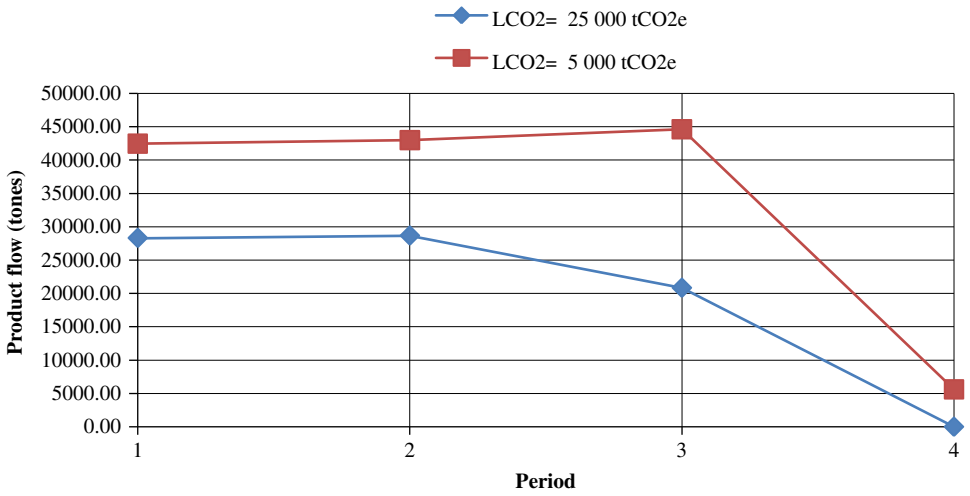


Fig. 10. Product flow between recycling centers and production units.

compliance to the regulation. Moreover, recycled products are less. This means that an environmental regulation that impose limits on GHG emission might lead to decrease in recycling activities if recycling costs are not optimized. Fig. 10 shows that the recycling activity is reduced under policy stringency and when carbon prices are very high.

7. Conclusion

In this article, we present a generic mathematical model to assist decision makers in designing sustainable supply chains over their entire life cycle. First, the model has the potential to be a tool that facilitates the understanding of optimal supply chain strategies under different environmental policies: recycling and GHG emissions reduction. The model shows that the various environmental legislations must be strengthened and harmonized at a global level in order to drive a meaningful long-term environmental strategy.

The explicit consideration of environmental costs within supply chain design is critical under the emergence of emission trading schemes. The integration of Life Cycle Analysis principles at the supply chain design phase maximizes the long-term sustainability. While some specific values of model's parameters would depend upon the application, the methodology presented here is general enough and may be applied to other supply chain studies to evaluate their performance in term of cost and carbon emissions.

Finally, although only the economic and environmental dimensions of sustainability are considered in the mathematical model, the methodology can integrate the social dimension as soon as measures of social stainability are identified.

Appendix

Different parameters are necessary to describe the model.

7.1. Monetary parameters

a^l_{nz} Fixed cost for locating site $n \in F \cup D \cup R$ at zone $z \in Z$.

a^s_{pnt} Fixed cost due to acquisition of raw material $p \in P^{MP}$ from supplier $n \in S$ during period $t \in T$. This represents the cost of development of long-term partnership with the supplier to guarantee a good service level (e.g. investing on information technology and communication).

c^{af}_{pnt} Purchasing cost of one unit of raw material $p \in P^{MP}$ from supplier $n \in S$ during time period $t \in T$.

c^{ar}_{pnt} Purchasing cost of one unit of recycled material $p \in P^{MP} \cup P^{PF}$ from recycling center $n \in R$ during time period $t \in T$.

a^f_{nt} Fixed cost associated with the operation of production unit $n \in F$ during time period $t \in T$.

a^h_{hnt} Fixed cost associated with the implementation of a new technology $h \in H$ in production unit $n \in F$ during time period $t \in T$.

a^p_{phnt} Production cost of one unit of product $p \in P^{PF}$ using technology $h \in H$ in production unit $n \in F$ during time period $t \in T$.

c^p_{phntz} Variable cost associated with the configuration of technology $h \in H$ used in production unit $n \in F$ to manufacture/assemble product $p \in P^{PF}$ during time period $t \in T$ at zone $z \in Z$.

c^{fi}_{pnt} Inventory carrying cost of input product $p \in P^{MP} \cup P^{PF}$ at production unit $n \in F$ during time period $t \in T$.

c^{fo}_{pnt} Inventory carrying cost of output product $p \in P^{PF}$ at production unit $n \in F$ during time period $t \in T$.

a^d_{nt} Fixed cost associated with the operation of distribution center $n \in D$ during time period $t \in T$.

c^d_{pnt} Inventory carrying cost of product $p \in P^{PF}$ at distribution center $n \in D$ during time period $t \in T$.

a^r_{nt} Fixed cost associated with the operation of recycling center $n \in R$ during time period $t \in T$.

c^{rc}_{pnt} Purchasing cost of one unit of used product $p \in P^{PF}$ at recycling center $n \in R$ during time period $t \in T$.

c^r_{pntz} Refurbishing cost of one unit of used product $p \in P^{PF}$ at recycling center $n \in R$ during time period $t \in T$ at zone $z \in Z$.

c^{dr}_{pntz} Disposal cost of used product $p \in P^{PF}$ at recycling center $n \in R$ during time period $t \in T$ at zone $z \in Z$.

c^{ri}_{pnt} Inventory carrying cost of product $p \in P^{PF}$ at recycling center $n \in R$ during time period $t \in T$.

c^{ro}_{pnt} Inventory carrying cost of raw material $p \in P^{MP}$ at recycling center $n \in R$ during time period.

$a^t_{nn'mt}$ Fixed cost associated with the establishment of a transportation link between node $n \in N$ and node $n' \in N$ using mode $m \in M$ during time period $t \in T$.

$c^t_{pnn'mt}$ Transportation cost of one unit of product $p \in P^{MP} \cup P^{PF}$ between node $n \in N$ and node $n' \in N$ using transportation mode $m \in M$ during time period $t \in T$

v_{it} Usage cost of input $i \in I$ during time period $t \in T$.

u_{ot} Emission cost of output $o \in O$ during time period $t \in T$.

A_{tz}^{cc} The market price of buying a credit during time period $t \in T$ at zone $z \in Z$.

V_{tz}^{cc} The market price of selling a credit during time period $t \in T$ at zone $z \in Z$.

7.2. Technical parameters

CF_{iph}^{p} Utilization factor of input $i \in I$ to manufacture/assemble $p \in P^{PF}$ using technology $h \in H$.

CF_{ipn}^{r} Utilization factor of input $i \in I$ during the recycling process of product $p \in P^{PF}$ at recycling center $n \in R$.

CF_{ipn}^{d} Utilization factor of input $i \in I$ during the destruction process of product $p \in P^{PF}$ at recycling center $n \in R$.

CF_{im}^{t} Utilization factor of input $i \in I$ for using transportation mode $m \in M$.

EF_{oph}^{p} Emission inventory of output $o \in O$ to manufacture/assemble product $p \in P^{PF}$ using technology $h \in H$.

EF_{opn}^{r} Emission inventory of output $o \in O$ during the recycling process of product $p \in P^{PF}$ at recycling center $n \in R$.

EF_{opn}^{d} Emission inventory of output $o \in O$ during the destruction process of product $p \in P^{PF}$ at recycling center $n \in R$.

EF_{om}^{t} Emission inventory of output $o \in O$ for using transportation mode $m \in M$

λ_{pnt} Capacity of supplier $n \in S$ for raw material $p \in P^{MP}$ during time period $t \in T$.

$\varphi_{pp'}$ Utilization factor of product $p \in P^{MP} \cup P^{PF}$ used in product $p' \in P^{PF}$.

q_{phnt}^{p} Upper bound on product $p \in P^{PF}$manufactured/assembled using technology $h \in H$ at production unit $n \in F$ during time period $t \in T$.

i_{pnt}^{fi} Inventory capacity for input product $p \in P^{MP} \cup P^{PF}$ at site $n \in F$ during period $t \in T$.

i_{pnt}^{fo} Inventory capacity for output product $p \in P^{PF}$ at node $n \in F$ during period $t \in T$.

i_{pnt}^{d} Inventory capacity for final products $p \in P^{PF}$ at node $n \in D$ during period $t \in T$.

χ_{nt} Distribution capacity at distribution center $n \in D$ during time period $t \in T$.

d_{pnt} Demand of product $p \in P^{PF}$ for customer $n \in C$ during time period $t \in T$.

δ_{pt} Return rate target for product $p \in P^{PF}$ during time period $t \in T$.

i_{pnt}^{ri} Inventory capacity of recovered products $p \in P^{PF}$ at recycling centers $n \in R$ during period $t \in T$.

θ_{pntz} Recycling performance factor for product $p \in P^{PF}$ at recycling center $n \in R$ during time period $t \in T$ at zone $z \in Z$.

$\phi_{pp'}$ Conversion factor of recycled material $p \in P^{MP} \cup P^{PF}$ from product $p' \in P^{PF}$.

q_{pnt}^{r} Capacity of recycling center $n \in R$ for product $p \in P^{PF}$ during time period $t \in T$.

$c_{nn'mt}$ Capacity of transportation between node $n \in N$ and node $n' \in N$ when using transportation mode $m \in M$ during time period $t \in T$.

7.3. Carbon management parameters

CO_{i}^{in} Characterization factor used to convert input $i \in I$ to carbon dioxide equivalent (CO_2e).

CO_{o}^{out} Characterization factor used to convert output $o \in O$ to carbon dioxide equivalent (CO_2e).

L_{tz}^{COP} Limit on the number of credits to purchase during compliance time period $t \in T$ at zone $z \in Z$.

L_{tz}^{COS} Limit on the number of credits to sold during compliance time period $t \in T$ at zone $z \in Z$.

$L_{tz}^{CO_2e}$ Aggregated limit in term of carbon dioxide equivalent (CO_2e) emissions during compliance time period $t \in T$ at zone $z \in Z$.

References

Barker, T.J., Zabinsky, Z.B., 2008. Reverse logistics network design: a conceptual framework for decision making. International Journal of Sustainable Engineering 1 (4), 250–260.

Bayon, R., Hawn, A., et al., 2007. Voluntary carbon markets: an international business guide to what they are and how they work. Sterling, Va., Earthscan, London.

Brentrup, F., Küsters, J., et al., 2001. Application of the life cycle assessment methodology to agricultural production: an example of sugar beet production with different forms of nitrogen fertilisers. European Journal of Agronomy 14 (3), 221–233.

Contreras, A.M., Rosa, E., et al., 2009. Comparative life cycle assessment of four alternatives for using by-products of cane sugar production. Journal of Cleaner Production 17 (8), 772–779.

De Benedetto, L., Klemes, J., 2009. The environmental performance strategy map: an integrated LCA approach to support the strategic decision-making process. Journal of Cleaner Production 17 (10), 900–906.

Dotoli, M., Fanti, M.P., et al., 2005. A multi-level approach for network design of integrated supply chains. International Journal of Production Research 43 (20), 4267–4287.

Dotoli, M., Fanti, M.P., et al., 2006. Design and optimization of integrated e-supply chain for agile and environmentally conscious manufacturing. IEEE Transactions on Systems, Man, and Cybernetics Part A:Systems and Humans 36 (1), 62–75.

Ferretti, I., Zanoni, S., et al., 2007. Greening the aluminium supply chain. International Journal of Production Economics 108 (1-2), 236–245.

Frota Neto, J.Q., Bloemhof-Ruwaard, J.M., et al., 2008. Designing and evaluating sustainable logistics networks. International Journal of Production Economics 111 (2), 195–208.

Goetschalcks, M., Fleischmann, B., 2008. Strategic network design. Supply Chain Management and Advanced Planning, 117–132.

Guillen-Gosalbez, G., Grossmann, I.E., 2009. Optimal design and planning of sustainable chemical supply chains under uncertainty. AIChE Journal 55 (1), 99–121.

Gunasekaran, A., Patel, C., et al., 2004. A framework for supply chain performance measurement. International Journal of Production Economics 87 (3), 333–347.

Hugo, A., Pistikopoulos, E.N., 2005. Environmentally conscious long-range planning and design of supply chain networks. Journal of Cleaner Production: Recent advances in Industrial Process Optimisation 13 (15), 1471–1491.

Jayaraman, V., 2006. Production planning for closed-loop supply chains with product recovery and reuse: an analytical approach. International Journal of Production Research 44 (5), 981–998.

Jayaraman, V., Guide Jr., V.D.R., et al., 1999. Closed-loop logistics model for remanufacturing. Journal of the Operational Research Society 50 (5), 497–508.

Johnson, E., Heinen, R., 2004. Carbon trading: time for industry involvement. Environment International 30 (2), 279–288.

Labatt, S., White, R.R., 2007. Carbon finance: the financial implications of climate change. John Wiley & Sons, Hoboken, N.J.

Lieckens, K., Vandaele, N., 2007. Reverse logistics network design with stochastic lead times. Computers and Operations Research 34 (2), 395–416.

Lu, L.Y.Y., Wu, C.H., et al., 2007. Environmental principles applicable to green supplier evaluation by using multi-objective decision analysis. Sustainable Design and Manufacture, International Journal of Production Research 45 (18-19), 4317–4331.

Luo, Y., Wirojanagud, P., et al., 2001. Comparison of major environmental performance metrics and their application to typical electronic products. In: Proceedings of the 2001 IEEE international symposium on electronics and the environment, May 7–9 2001. Denver, CA, Institute of Electrical and Electronics Engineers Inc., pp. 94–99.

Luo, Y., Zhou, M., et al., 2001. An integrated E-supply chain model for agile and environmentally conscious manufacturing. IEEE/ASME Transactions on Mechatronics 6 (4), 377–386.

Min, H., Melachrinoudis, E., 1999. The relocation of a hybrid manufacturing/distribution facility from supply chain perspectives: a case study. Omega 27 (1), 75–85.

Nagurney, A., Liu, Z., et al., 2006. Optimal endogenous carbon taxes for electric power supply chains with power plants. Mathematical and Computer Modelling 44 (9–10), 899–916.

Nagurney, A., Toyasaki, F., 2003. Supply chain supernetworks and environmental criteria. Transportation Research Part D: Transport and Environment 8 (3), 185–213.

Peace, J., Juliani, T., 2009. The coming carbon market and its impact on the American economy. Policy and Society 27 (4), 305–316.

Pistikopoulos, E.N., Hugo, A., 2005. Environmentally conscious long-range planning and design of supply chain networks. Journal of Cleaner Production 13 (15), 1428–1448.

Pochampally, K.K., Nukala, S., et al., 2009. Strategic planning models for reverse and closed-loop supply chains, vol. xvii. CRC Press, Boca Raton 273p.

Ramudhin, A., Chaabane, A., et al., 2008. Carbon market sensitive green supply chain network design. In: Proceedings of the IEEE international conference on industrial engineering and engineering management, 2008. IEEM 2008.
Ramudhin, A., Chaabane, A., et al., 2010. Carbon market sensitive sustainable supply chain network design. International Journal of Management Science and Engineering Management 5 (1), 30–38.
Rice, G., Clift, R., et al., 1997. Comparison of currently available european LCA software. The International Journal of Life Cycle Assessment 2 (1), 53–59.
Sabri, E.H., Beamon, B.M., 2000. A multi-objective approach to simultaneous strategic and operational planning in supply chain design. Omega 28 (5), 581–598.
Schultmann, F., Zumkeller, M., et al., 2006. Modeling reverse logistic tasks within closed-loop supply chains: an example from the automotive industry. European Journal of Operational Research 171 (3), 1033–1050.
Seuring, S., Muller, M., 2008. Core issues in sustainable supply chain management—a Delphi study. Business Strategy and the Environment 17 (8), 455–466.
Sheu, J.-B., 2008. Green supply chain management, reverse logistics and nuclear power generation. Transportation Research Part E: Logistics and Transportation Review 44 (1), 19–46.
Sheu, J.-B., Chou, Y.-H., et al., 2005. An integrated logistics operational model for green-supply chain management. Transportation Research Part E: Logistics and Transportation Review 41 (4), 287–313.
Srivastava, S.K., 2008. Network design for reverse logistics. Omega 36 (4), 535–548.
Stranlund, J., 2007. The regulatory choice of noncompliance in emissions trading programs. Environmental and Resource Economics 38 (1), 99–117.
Subramanian, R., Talbot, F.B., et al., 2008. An approach to integrating environmental considerations within managerial decision-making, SSRN eLibrary http://ssrn.com/paper=1004339.
White, C.D., Masanet, E., et al., 2003. Product recovery with some byte: an overview of management challenges and environmental consequences in reverse manufacturing for the computer industry. Journal of Cleaner Production 11 (4), 445–458.
Zhou, Z., S. Cheng, et al., 2000. Supply chain optimization of continuous process industries with sustainability considerations. In: Proceedings of the 7th international symposium on process systems engineering, 16–21 July 2000, Keystone, CO, USA, Elsevier.